凌辰(Chen Ling)

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工作经历

Technical Expert,	紫光前沿院	2025.8-
Technical Expert,	英特尔 Mobileye	2022.6 – 2025.3
Staff Engineer(Grade 8),	英特尔	2017.10 – 2022.6
Sr. Android Engineer,	瞬联科技	2014.11 – 2017.10
Algorithm Engineer,	汉王科技	2010.3 – 2014.10

个人优势

- 拥有多年深度学习、大语言模型（LLM， MoE，LlamaFactory）、计算机视觉及自动驾驶领域的经验
- 具备端到端模型流程设计管理经验(数据，训练，推理，量化，剪枝)
- 具备 GPU/NPU (CUDA/Cutlass) 优化和加速执行流程的经验(GEMM, 归一化，SIMD & SIMT)
- 具备构建中等规模 GPU 集群以及高参数量模型集群训练优化经验(张量并行，模型&流水线并行)
- 擅长项目管理，曾领导多个复杂项目开发以及多模块代码库重构
- 多年世界 500 强公司 R&D 经历,英语流利，可以使用英语作为工作汇报语言，熟悉跨国以及跨时区项目工作
- 良好的人际交往能力—擅长沟通和辅助新人成长，具备技术领导力和团队合作精神
- 工作富有创新精神、自学能力强，能够快速掌握新技术。

项目经历

关键领域： 多模态大模型 (MLLM)，深度学习， CUDA, AI Agent, 机器学习，计算视觉，训练集群构建, Transformer, Python, C++, Pytorch, Tensorflow,

PyTorch 算子支持与推理引擎适配(紫光前沿院）｜ 2025-

技术专家

- 主要负责 SGLang 整体平台支持，包括 Radix Attention, 多进程通信，PD 分离，MoE，PyTorch 算子支持以及多个 LLM 模型适配，包括 Qwen2 VL, Qwen3, Llama 3.2 等

基于 AI Agent 的自动化车载 TSR-OCR 系统研发平台（英特尔 Mobileye）｜ 2024-2025

项目负责人

- 主导设计并构建了基于大语言模型（LLM）的 AI Agent 自动化工作流，将交通标识（TSR-OCR）模型的完整生命周期（数据处理、模型训练（TensorFlow）、评估、打包部署）整合为端到端自动化流程。**成功将新模型迭代与发布周期从 16 周缩短至 4 周，效率提升 75%。**
- 构建小型分布式训练集群(4 Machines, 32 GPUs, 40Gbps)，综合运用 PyTorch+DeepSpeed+Megatron，设计并实施张量、流水线并行（TP/PP）的混合(3D 并行)策略，高效完成模型的有监督微调（SFT），并探索 LoRA 与量化（AWQ）技术进行进一步优化。引入强化学习（PPO/DPO）对 Agent 的决策逻辑进行训练，并探索 MoE 架构以提升其处理复杂任务的效率。
- 负责领域专用多模态大模型（Qwen2 VL 72B/7B）的优化与微调，以驱动 AI Agent 的核心能力。
- 构建并部署了高性能 LLM 推理服务，为 AI Agent 提供实时决策能力。部署 vLLM 框架并集成 Paged Attention，同时探索投机解码（Speculative Decoding）等先进解码策略，最大化推理吞吐量与响应速度。
- 利用多模态 LLM 的能力重塑数据处理流程，将传统劳动密集型的标注与审核环节智能化。通过该系统，数据处理效率从 9 周缩减至 3 周，提升超过 60%。

车载 NPU 芯片部署与算子研究（英特尔 Mobileye）｜ 2022-2025

项目负责人

- 引入 Reduce 方法加速归一化模块的推理以及在车载端部署。

- 研究注意力模型 Flash Attention 2 (Transformer) 和 PTX 指令级别部署，与以色列团队协作加速 EyeQ 芯片的推理。
- 应用 GEMM 分块加速、双缓冲、异步拷贝和计算密度优化进行预处理优化(CUDA/Cutlass)。
- 利用 AVX 以及 VMP 指令(SIMD)对图像进行预处理包括图像缩放以及直方图操作
- 对算子融合进行研究，补齐片上系统(SoC)算子水平及垂直融合短板

基于扩散模型（Diffusion）生成数据的 DMS 驾驶员监控系统（英特尔 Mobileye）| 2022-2023

数据及感知负责人

- 主导人脸检测及识别模块（改进 YoloX/RetinaNet/RepVGG），实现推理速度翻倍，满足芯片性能要求。
- 通过 Face Mesh & Prompt，改进引导扩散模型（ControlNet、ReferenceNet），加入可编辑引导生成流程，并集成进人脸数据（红外和 RGB 图像）标注流程，将训练集数据量扩展 4 倍，同时提升模型精度 5%。
- 改进 Paint-by-example 和 StyleGAN2，为 Face 示例添加口罩&胡须、改变眼睛开闭状态，并应用于面部状态识别模型，提升识别精度约 6%。
- 成功领导代码库(60000+行，15+模块）重构，使用 Confluence + teams + Jira + GitLab 对项目重构进行管理。

离线道路数据验证（英特尔 Mobileye）| 2022

团队负责人

- 带领整个验证团队推动合规室建设，按照中国数据安全法审核数据，并在北京 Site 构建离线数据验证流程（云端结果验证），完成中国道路数据的 TSR、AEB、HLB 模块验证。

多相机多目标跟踪与场景重建（英特尔）| 2017-2022

数据团队负责人，基础设施团队负责人，项目负责人，

- 改进 YOLO(小物体特征图，NMS)并训练检测模块以适配 18 台相机的 5k 分辨率图像实时推理优化（CUDA 优化），同时保持精度，并通过 TensorRT 和 OpenVINO 使用 FP16 和 INT8 量化加速检测模块的推理。
- 主导 NFL 球员跟踪流程，将精度从 69% 提升至 97%，使用 CNN 多视图深度估计算法（MVSnet）重建高光时刻，根据多路相机模型，实时计算球员 BEV 位置及轨迹信息。
- 设计多阶段验证的 CI/CD 流程，将整个验证数据集处理速度提升约 20 倍，构建分布式训练集群（16 台机器，每台配备 8 张 2080ti 显卡+NCCL），通过 DDP 线性加速训练过程。
- 全面负责数据团队，构建数据获取，标注以及工作流中的中间数据分析流程

Pocket Avatar Apps (CIeNet) | 2014-2017

高级算法工程师

- 利用 PnP 方法实现 3D 虚拟饰品与前置摄像头人脸融合应用(OpenCV)，并利用 OpenGL 计算内核加速传统图像算法（如卷积、模糊、Canny 边缘检测和变形）。
- 开发**跨平台**（iOS、Android 和 Windows）的 Pocket avatar 演示程序。

绘图软件 Coolpaint (Hanvon) | 2010-2014

算法工程师，项目负责人

- 使用 OpenCL 和 CUDA 加速传统图像算法，确保不同显卡（AMD-ATI、Nvidia）平台上的执行结果一致。
- 提高绘画软件的模拟真实感，模拟 Photoshop 实现轨迹叠加、油画笔等效果，并使用非真实感算法实现轨迹混色效果。

专利情况

作为第一发明人申请 3 项专利，参与 7 项专利，部分如下：

- 3D 运动场景中球员跟踪的高光时刻识别
- 大场景多相机多目标跟踪中带检测模块的球衣识别

获奖情况

- 英特尔 “最佳专利团队”
- Kaggle 分割竞赛（Carvana），银牌（前 4%）Kaggle 分类竞赛（C-discount），铜牌（前 12%）

教育背景

硕士，计算机软件与理论 天津理工大学(计算机学科评估 B) *2007-2010*

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WORKING EXPERIENCE

Technical Expert,	Omnicom Tech	2025 –
Technical Expert,	Intel Mobileye	2022 – 2025
Staff Engineer(Grade 8)	Intel	2017 – 2022
Sr. Android Engineer,	CIeNET Technologies	2014 – 2017
Algorithm Engineer,	Hanvon Technologies	2010 – 2014

STRENGTH

- Years of experience in large language models (LLM, MoE, Decoding, LlamaFactory), deep learning (MLops), computer vision, and autonomous driving.
- Expertise in end-to-end model workflow design and management (data processing, training, inference, quantization, pruning).
- Hands-on experience in GPU/NPU (CUDA/Cutlass) optimization and execution acceleration (GEMM, normalization, SIMD & SIMT parallel computing).
- Proficiency in building medium-scale GPU clusters and optimizing cluster training for high-parameter models (tensor parallelism, model & pipeline parallelism).
- Good at project management, led the development of multiple complex projects and the refactoring of multi-module repo.
- Fluent in English, capable of using English as the working report language, and familiar with cross-border and cross-time zone project work.
- Good interpersonal skills—excellent at communication and assisting new team members in growth, with technical leadership and team cooperation spirit.
- Working with an innovative spirit and strong self-learning ability, able to quickly master new technologies.

PROJECTS

Key Areas: Multimodality Large Language Models (MLLM), Deep Learning, Computer Vision, CUDA, Training Cluster Establishment, Transformer, Python, C++, Pytorch, Tensorflow

PyTorch Operator Support and Inference Engine Adaptation (Omnicom Tech) | 2025-

Tech Expert

- Primarily responsible for the overall platform support of SGLang, including Radix Attention, multi-process communication, PD separation, MoE, PyTorch operator support, and adaptation of multiple LLM models such as Qwen2 VL, Qwen3, and Llama 3.2.

LLM-Centric R&D Platform for Automotive TSR-OCR Automation (Intel Mobileye) | 2024-2025

Project Leader

- Architected and built an AI Agent-driven automated workflow based on (LLMs) to automate the end-to-end lifecycle of Traffic Sign Recognition (TSR-OCR) models, including data processing, training, evaluation, and deployment. This successfully reduced **the model iteration and release cycle from 16 weeks to 4 weeks, a 75% efficiency improvement.**
- Constructed a small-scale distributed training cluster (4 machines, 32 GPUs, 40Gbps), leveraging PyTorch + DeepSpeed + Megatron to design and implement a hybrid 3D parallel strategy (tensor and pipeline parallelism, TP/PP). Efficiently executed supervised fine-tuning (SFT) of models, and explored LoRA and quantization (AWQ) for further optimization. Introduced reinforcement learning (PPO/DPO) to train the agent's decision-making logic and investigated MoE architecture to enhance its efficiency in handling complex tasks.
- Oversaw the optimization and fine-tuning of domain-specific multimodal large models (Qwen2 VL 72B/7B) to drive the core capabilities of AI agents.
- Built and deployed high-performance LLM inference services to enable real-time decision-making for AI agents. Deployed the vLLM framework（Paged Attention, PD disaggregation) and explored advanced decoding strategies such as speculative decoding to maximize inference throughput and response speed.
- Re-engineered the data processing pipeline by leveraging the multimodal LLM's capabilities, automating the traditionally labor-intensive annotation and review stages. **This initiative cut data processing time from 9 weeks to 3 weeks, an efficiency gain of over 60%.**

On-Board NPU Chip Deployment and Operator Research (Intel Mobileye) | 2022-2025

Project Leader

- Introduced reduced methods (down & xor) to speed up the inference of the normalization module.
- Researched on **Flash Attention 2(Transformer) and PTX migration** to collaborate Israeli colleagues to expedite inference on EyeQ.
- Applied GEMM tiling acceleration, double buffering, async copy and arithmetic density improvement for pre-processing optimization. (**CUDA/Cutlass**)
- Leveraged AVX & VMP instructions (**SIMD**) to do image pre-processing (scaling, histogram).
- Conducted research on operator fusion to address the shortcomings in vertical operator fusion for EyeQ SoC.

DMS Driver Monitoring System Relying on Diffusion-Generated Data (Intel Mobileye) | 2023-2024

Data & Perception Project Leader

- Led the development of face detection and recognition modules (improved YoloX/RetinaNet/RepVGG), doubling inference speed to meet chip performance requirements.
- Enhanced diffusion models (ControlNet, ReferenceNet) with Face Mesh & Prompt, introducing an editable guidance generation workflow. Integrated this into the annotation pipeline for facial data (infrared and RGB images), expanding the training dataset by 4x while improving model accuracy by 5%.
- Optimized Paint-by-example and StyleGAN2 to add masks/beards and modify eye states in face examples. Applied these enhancements to facial state recognition models, boosting accuracy by ~ 6%.
- Refactored the DMS repository (60000+lines, 15+module) successfully, using Confluence + teams + Jira + GitLab for the project refactoring management.

Offline Road Data Validation (Intel Mobileye) | 2022

Head of Offline-Validation Team

- Led whole validation team and drove the compliance room establishment to review the data follow the China Data Security Law and created the offline data validation process (Cloud results validation) in China Site to complete the TSR, AEB, HLB module validation on China Road data.

Multi-Camera Multi-Target Tracking and Scene Reconstruction (Intel) | 2017-2022

Head of Data Team, Head of Algorithm Infrastructure Team, Project Leader

- Improve YOLO (feature maps, NMS) and train detection-related modules to adapt to real-time inference optimization (CUDA optimization) for 5k resolution images from 18 cameras, while maintaining accuracy. Additionally, accelerate the inference of the detection module using FP16 and INT8 quantization through TensorRT and OpenVINO.
- Led the NFL player tracking procedure (Detector & Classifier & 3D projection & trajectory tracking), improved accuracy from 69% to 97%, reconstructed freezing moment with the multi-camera depth estimation algorithm.
- Designed the CI/CD for multi-stage validation to speed up the whole validation dataset by ~40x, constructed the training cluster (16 machines, each has 8 card 2080ti+**NCCL**) to speed up the training process linearly.
- Data Team Leader, constructing data acquisition, annotation, and intermediate data analysis processes in workflows.

Pocket Avatar Apps (CIeNet) | 2014-2017

Sr. Algorithm Engineer

- Utilized the PnP method to realize 3D virtual ornaments and front-camera face fusion applications and leveraged the OpenGL compute kernel accelerates image algorithms such as convolution, blur, canny, and warping.
- Developed for the **cross-platform** (iOS, Android and Windows) development of the Pocket Avatar Demo program

Painting software, Coolpaint (Hanvon) | 2010-2014

Algorithm Engineer, Project Leader

- Speed up the traditional image algorithms using OpenCL & CUDA and ensure consistent execution results across different graphics cards platforms. Increased the simulation realism of the painting software, emulated Photoshop to achieve track overlay, oil brush and other effects, and used non-realistic algorithms to achieve track color mixing effects.

PATENTS

3 patents were filed as the **first inventor,** 7 patents as participants**, part of them is as follows.**

Highlight freezing moment for player tracking in 3D movement scenario

Jersey Recognition with Detection module in Big Scene multi-camera multi-object tracking

HONORS

- Best Patents Team (Intel)
- Kaggle Segmentation Competition (Carvana), Silver (4%), Classification Competition (C-discount), Bronze (12%)

EDUCATION

M.S., Computer Software and Theory Tianjin University of Technology (Computer Science Discipline Evaluation: B) *2007-2010*